

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR I

ELEMENTARY BOTANY

UNIT II — External Morphology

DETAILED TEACHER-STYLE NOTES

Definitions • Every concept explained • Tables • Labelled study diagrams • Flowcharts • Examples • Comparisons • Exam points

Prepared in the same detailed format as Major Course II and Major Course III notes.

UNIT II — EXTERNAL MORPHOLOGY

The supplied roadmap places root, stem, leaf, flower, fruits and seeds under external morphology.

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1. Root

Definition — Root: Usually the descending, non-green plant organ that develops from the radicle and mainly performs anchorage and absorption of water and minerals.

Functions

- Anchorage in the soil.
- Absorption of water and mineral salts.
- Conduction toward the shoot.
- Storage in modified roots.
- Special functions such as support, respiration or climbing in some plants.

Types of root systems

Type	Origin and feature	Examples
Tap root	Primary root persists and gives lateral branches	Mustard, pea
Fibrous/adventitious	Many similar roots arise from stem base	Wheat, grasses
Adventitious root	Arises from parts other than the radicle	Banyan, maize, sweet potato

Root modifications

Modification	Purpose	Example
Storage	Stores reserve food	Carrot, radish, sweet potato
Prop roots	Extra mechanical support	Banyan
Stilt roots	Support from lower stem nodes	Maize
Pneumatophores	Gas exchange in waterlogged conditions	Mangrove plants
Climbing roots	Attachment to support	Money plant

2. Stem

Definition — Stem: The ascending axis of the plant that bears leaves, buds, branches and reproductive structures and acts as a major pathway for conduction.

Key features

- Nodes: points from which leaves or branches arise.
- Internodes: regions between two successive nodes.
- Terminal bud: may contribute to increase in length.
- Axillary buds: occur in leaf axils and may form branches or flowers.

Underground stem modifications

Type	Identification clue	Example	Main role
Rhizome	Horizontal stem with nodes/internodes	Ginger	Storage and perennation
Tuber	Swollen stem with buds ('eyes')	Potato	Storage and propagation
Corm	Short, vertical, swollen stem	Colocasia	Storage
Bulb	Reduced stem with fleshy scale leaves	Onion	Storage and perennation

3. Leaf

Definition — Leaf: A lateral, usually flattened organ borne at a node and specialized primarily for photosynthesis, transpiration and gas exchange.

Parts and characters

- Leaf base attaches the leaf to the stem.
- Petiole positions the lamina; a leaf may be sessile when petiole is absent.
- Lamina is the expanded photosynthetic part.
- Venation may be reticulate or parallel.
- Leaves may be simple or compound depending on division of the lamina.

Feature	Simple leaf	Compound leaf
Lamina	Undivided or not separated into leaflets	Divided into distinct leaflets
Axillary bud	Present at base of entire leaf	Not present at base of individual leaflet
Example	Mango	Neem

4. Flower

Definition — Flower: A specialized reproductive shoot bearing floral organs that participate in sexual reproduction.

Four major floral whorls



Ovary position

Term	Position idea	Typical example
Hypogynous	Other floral parts arise below a superior ovary	Mustard
Perigynous	Floral parts appear around the ovary on a cup-like thalamus	Rose (common textbook example)
Epigynous	Other floral parts appear above an inferior ovary	Guava

5. Fruits and seeds

After fertilization, the ovary commonly develops into a fruit and ovules commonly develop into seeds. A fruit may protect developing seeds and assist dispersal. Seed structure commonly includes seed coat, embryo and, in many cases, nutritive tissue.

Reproductive sequence



Exam focus: For morphology answers: definition → types → distinguishing characters → examples → labelled diagram. Do not mix root modifications with stem modifications.